Alternative QFT

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abc

Introduction.—

- We begin with the hidden subgroup problem (HSP) as a central source of exponential quantum speedups and use it to motivate why Fourier-type interference remains a foundational primitive in quantum algorithms.

- Standard HSP solvers rely heavily on the quantum Fourier transform (QFT), but implementing the full QFT still requires long-range controlled-phase operations and nontrivial circuit depth, which remain challenging in near-term experiments.

- Most existing approaches reduce the cost of QFT by constructing approximate-QFT circuits; however, this line of work still treats QFT itself as the target object and focuses mainly on gate-level simplification.

- Our perspective is different: instead of approximating the QFT gate by gate, we ask which structural properties of QFT are actually responsible for extracting hidden subgroup information.

- We focus on two such properties, namely shift invariance and Fisher information. Shift invariance characterizes whether the interference pattern induced by subgroup cosets is preserved in the measurement statistics, while Fisher information quantifies how efficiently the hidden subgroup parameter can be inferred from samples.

- Based on this perspective, we identify a family of circuits that satisfies the shift-invariance requirement and therefore serves as a principled alternative to QFT rather than a purely heuristic approximation.

- We then numerically study how the Fisher information of this circuit family scales with the number of qubits and with circuit depth, which provides evidence about whether the hidden subgroup can still be recovered with feasible classical post-processing.

- Our experiments indicate that the Fisher-information scaling remains acceptable over the parameter regimes we test, suggesting that even relatively shallow circuits may retain enough information for hidden subgroup recovery.

- To make this claim operational, we use a neural network as a classical post-processing module and show that the hidden subgroup can indeed be reconstructed from the corresponding measurement data.

- Finally, we compare the behavior of the proposed circuits in the presence of noise, with the goal of understanding when they may offer a practical advantage over standard QFT-based implementations.

Shift invariant circuit.—

- We first recall the mathematical definition of HSP and the standard quantum routine used to solve it, so that the role played by Fourier-type transformations is stated precisely.

- Within this routine, we explain why shift invariance is the key property that allows different cosets of the hidden subgroup to interfere in a controlled and information-preserving way.

- We then introduce our formal definition of a shift-invariant circuit and discuss how it captures the operational requirement needed by the HSP measurement stage.

- Starting from this definition, we derive necessary conditions that any candidate circuit must satisfy, which narrows the search space for QFT alternatives and clarifies the structure behind successful constructions.

PH circuit.—

- We identify a concrete circuit family that satisfies the shift-invariance conditions derived above, thereby providing an explicit nontrivial replacement for the usual QFT circuit.

- This family is strictly broader than the set generated by standard QFT constructions, which shows that shift invariance is not unique to QFT and can arise in a larger design space of useful circuits.

HSP solver for $\mathbb{Z}_{2^n}$.—

- We first study the case of the cyclic group $\mathbb{Z}_{2^n}$, because it is naturally encoded by an n-qubit register and therefore provides the cleanest setting for circuit-level analysis.

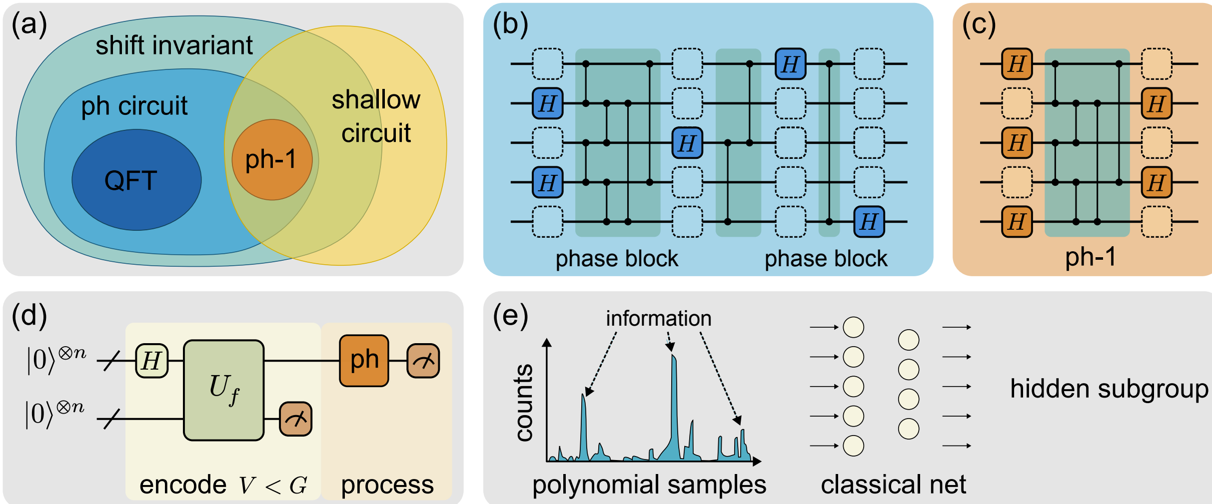


FIG. 1. 1

- In this setting, the subgroup structure is especially transparent, which allows us to isolate the role of the circuit transformation from complications caused by group embedding or discretization.
- We show that for this family of groups, a transversal layer of Hadamard gates already captures the information needed to identify the hidden subgroup in the idealized setting.
- This gives a concrete algorithmic baseline and helps explain why shallow, structured alternatives to QFT may already be sufficient for a meaningful subclass of HSP instances.

Approximate $\mathbb{Z}_q$.—

- The previous setting is mathematically clean, but it is also algorithmically limited because classical methods are already effective in several such special cases, reducing the practical need for a quantum advantage.
- The more interesting regime is to work over $\mathbb{Z}$ and recover the subgroup $\mathbb{Z}_q$ through a finite-dimensional approximation, which is where the standard QFT demonstrates its real power.
- A key strength of QFT is that it can use a sufficiently large register $\mathbb{Z}_{2^n}$ to approximate the infinite-group problem and still recover $\mathbb{Z}_q$, even when $\mathbb{Z}_q$ is not literally a subgroup of $\mathbb{Z}_{2^n}$.
- We therefore ask whether our shift-invariant circuit family can inherit part of this robustness, and in particular whether some shallow members of the family remain useful in this approximate setting.

Neural net as post processing.—

- We use a neural network as a learnable classical post-processing layer that takes measurement outcomes from the candidate circuit family and predicts the hidden subgroup parameter.
- This experiment is intended to test the practical recoverability of subgroup information, rather than only reporting information-theoretic quantities in the abstract.
- The resulting performance shows that the measurement distributions produced by these circuits still contain usable structure, and that a flexible classical decoder can successfully exploit it.

Fisher information.—

- To justify the post-processing approach more systematically, we ask whether the hidden subgroup parameter can be estimated using only a polynomial number of measurement samples.
- Fisher information provides a natural metric for this question because it quantifies the local statistical sensitivity of the measurement distribution to changes in the hidden parameter.
- We therefore compute the Fisher information numerically as a function of circuit depth and system size, and use these trends to assess whether efficient recovery should remain possible.
- The observed scaling offers evidence that the proposed circuit family can preserve enough information for practical inference, even when the circuit is considerably shallower than a full QFT implementation.

PH performance under noise—

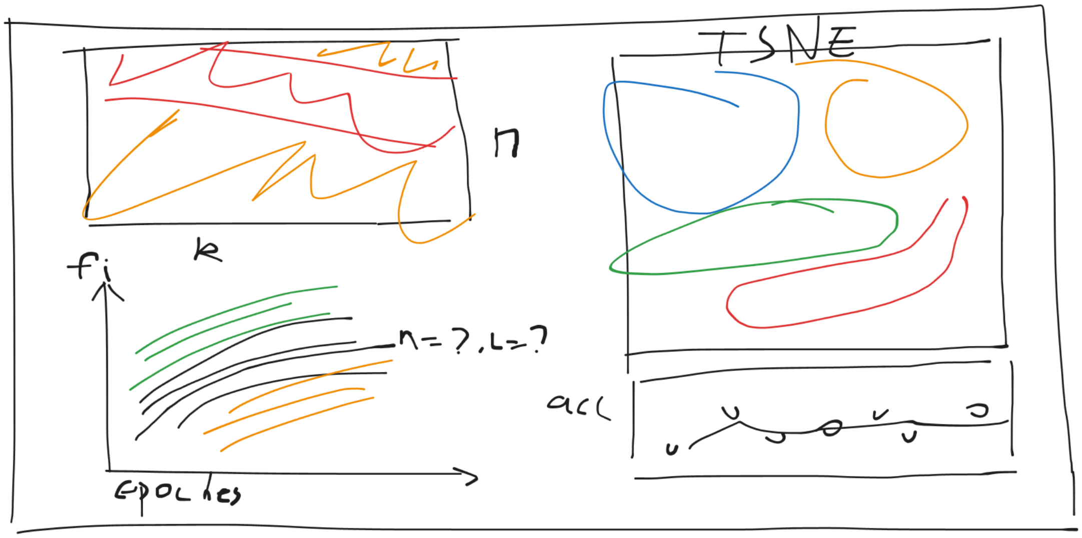


FIG. 2. 2

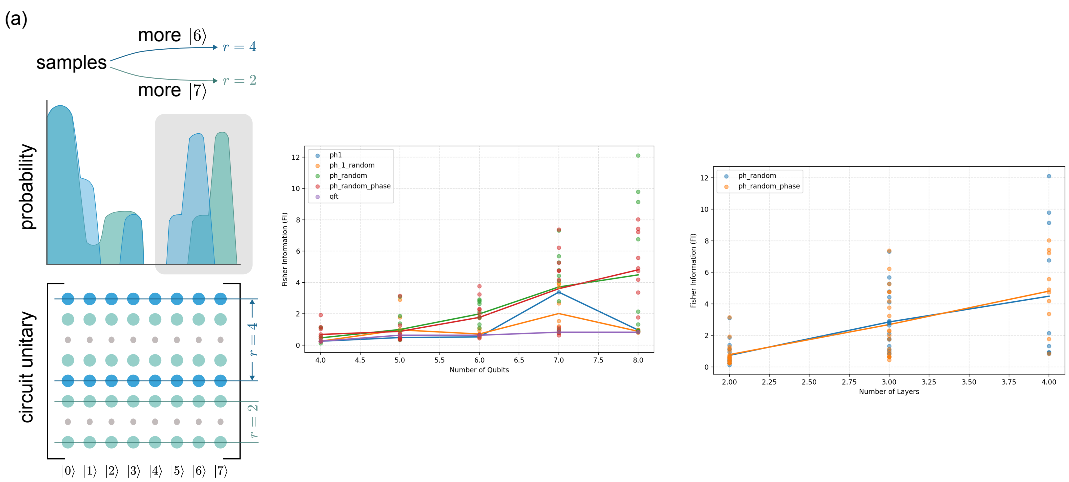
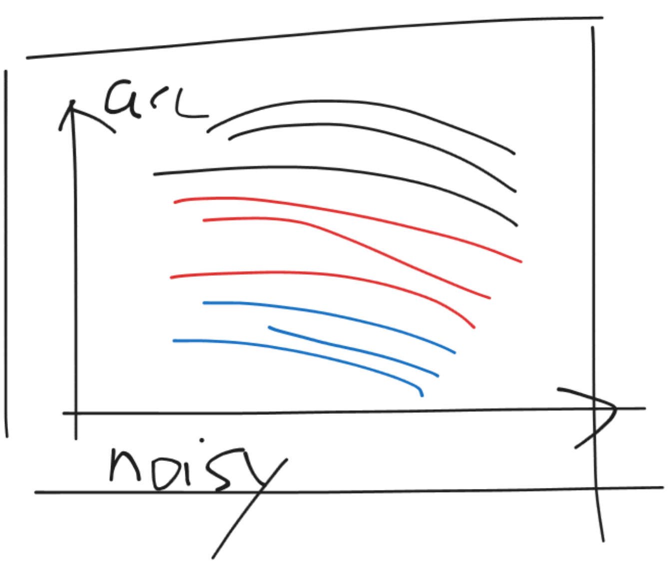


FIG. 3. 2



- We evaluate the proposed circuit family under noisy dynamics to determine whether its structural simplifications also translate into improved robustness in realistic hardware settings.

- In particular, we compare its degradation pattern with that of standard QFT-based circuits and analyze whether shallower alternatives can preserve useful subgroup information for longer.

Shift invariant circuit

Theorem 1. *A shift invariant circuit* U *over abelian group* G *satisfy the condition* $U_{ij} = 1/\sqrt{|G|}e^{\mathrm{i}\theta}$.

Proof. 123 $\square$

PH circuit is shift invariant